Models of mother-child attachment

"I wanna hold your haaaaaaand."

L. F. Rossi

Graduate Student Seminar

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Collaborators

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Outline

1. Problem statement and description
2. Game theoretic approach
   - A simple single payoff problem
   - A more complex single payoff problem
   - A double payoff problem

A strange situation

1. Mother and child enter a new room with toys. The child begins to play.
2. A stranger enters the room and speaks with the mother.
3. The mother leaves the room. (Child anxiety rises.)
4. After three minutes, the mother returns to the room. The child responds.

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Observed modes of child behavior

Secure child

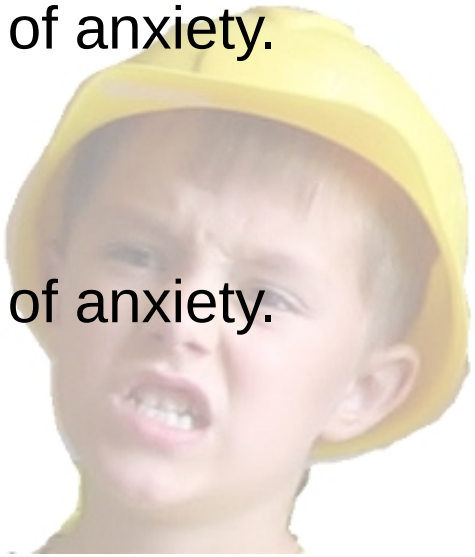
- Receives from mother quick and total down regulation of anxiety.
- High mobility.

Ambivalent child

- Receives from mother partial down regulation of anxiety.
- Low mobility (close to mother, clingy).

Avoidant child

- Receives from mother partial down regulation of anxiety.
- Low mobility (far from mother).



Other approaches

- Dynamical model: Variables for position, emotional state and other internal drivers. Lots of parameters.
- Control theory model: Gates, integrators and differentiators. Lots of parameters.

Decision making models (game theory)

Model the behavior of the child and the mother as a decision making process where each participant is trying to optimize its own situation.

- Each player implements a strategy. In our case, we assume a strategy is a list of probabilities.
- For each set of strategies, there is an expected payoff.
- Each player will try to choose the strategy that optimizes their own situation.
- The outcomes are associated with two experimental observables: Behavior (movement, crying, etc) and anxiety level.

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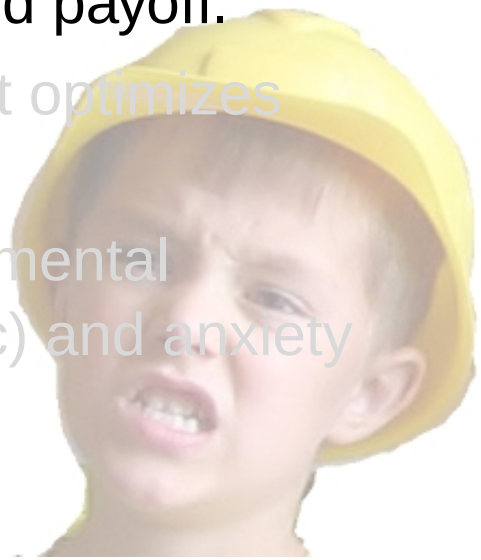

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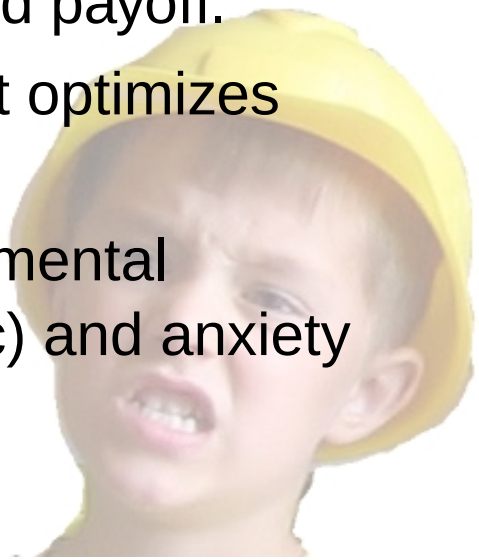

Terminology

- Payoff: The reward a player receives. Different players may have different payoff structures.
- Pure strategy: A strategy where the player always does the same thing.
- Mixed strategy: A strategy where the player makes different decisions either deterministically or stochastically.
- Zero-sum game: A game where the payoffs to all players sums to zero.
- Nash equilibrium: A situation where a change in strategy does not improve an individual's payoff.
- An equalizing strategy: A strategy giving your opponent the same payoff, no matter what the opponent does.

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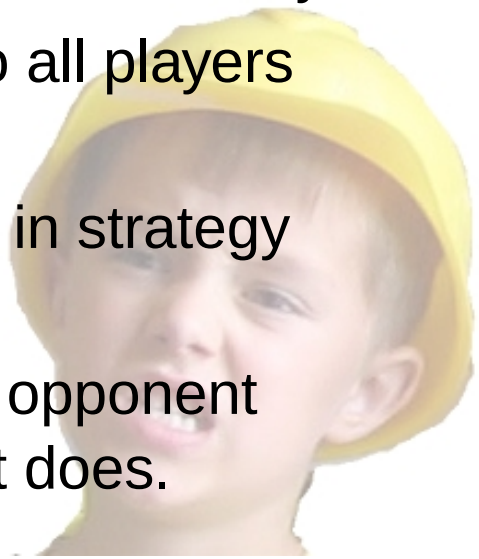
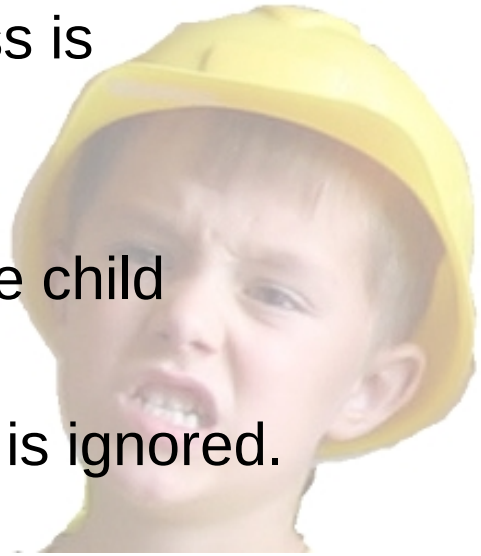


Single payoff

		Mother's action	
		Attend	Ignore
		q	1-q
Child's action	Go	1	-s
	Don't Go	0	0

The payoff is the amount by which the child's stress is down-regulated.
Two parameters:

- q: The probability of the mother comforting the child represents parenting behavior.
- $s > 0$: The increase in anxiety when the child is ignored.



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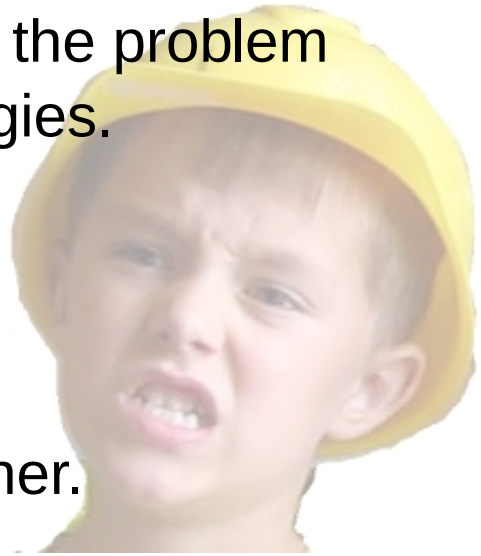
Analysis:
There is only one player and one payoff matrix, so the problem is linear. The optimal strategies will be pure strategies.

$$P_{\text{Go}} = q - (1-q)s,$$
$$P_{\text{No go}} = 0.$$



If $q > \frac{s}{s+1}$, the optimal strategy is to go to the mother.
If $q < \frac{s}{s+1}$, the optimal strategy is to stay put.

Single payoff

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Interpretation:

$$P_{\text{Go}} = q - (1-q)s,$$
$$P_{\text{No go}} = 0.$$

If $q > \frac{s}{s+1}$, the optimal strategy is to go to the mother. (Secure (large p) or ambivalent (smaller p))
If $q < \frac{s}{s+1}$, the optimal strategy is to stay put. (Avoidant.)

A richer decision problem

		Mother's action	
		Attend	Ignore
		q	1-q
Child's action	Go	1	-s
	Don't Go	0	0
	Half-Go	h	-t

Assume $u < t < s < h$.

In a high stress situation, we interpret "Half go" as a tentative approach to mother for comfort, with reduced expectation/willingness to accept comfort.

Child chooses row with optimal (i.e. lowest) payoff. The choice depends on (p_1, p_2).

E.g. If $p_1 + p_2 = 1$ (no frightening behaviour), there will be two thresholds p_A and p_B ($0 \leq p_A \leq p_B \leq 1$) such that

- child chooses "Don't go" if $p_1 < p_A$
- child chooses "Go" if $p_1 > p_B$
- child chooses "Half go" if $p_A < p_1 < p_B$

Mother and child play together

			Mother's action	
			Attend	Ignore
			q	$1-q$
Child's action	Go	p	$(1, 1-c)$	$(-s, -s)$
	Don't Go	$1-p$	$(0, -c)$	$(0, 0)$

Two parameters:

- $s > 0$: The increase in anxiety when the child is ignored.
- $c > 0$: The cost to the mother for providing comfort.

Mother and child play together: pure strategies

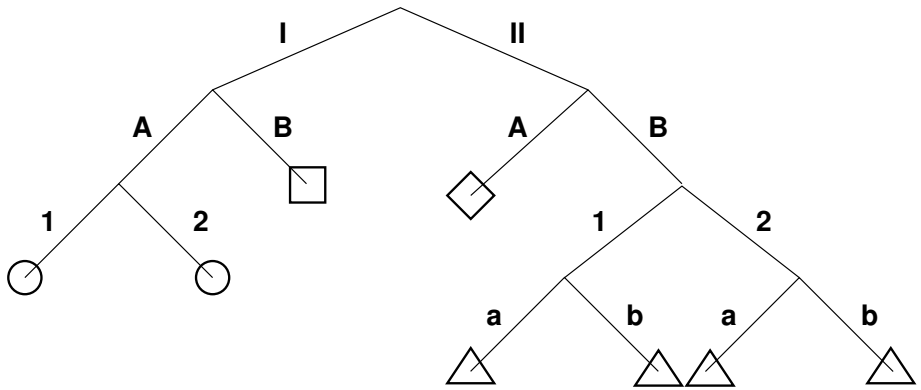
$$A = \begin{bmatrix} 1 & -s \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1-c & -s \\ -c & 0 \end{bmatrix}.$$

0<−s<1

1−c < −s

1−c < 0

−c < −s



0 > −s

1−c > −s

1−c > 0

−c > −s

When personalities meet

Group 1:

(4,2)	**(3,4)**
(2,1)	(2,3)

Type I A 1

(4,3)	**(3,4)**
(2,1)	(2,2)

Type I A 2

Group 2:

(4,4)	(3,3)
(2,1)	(2,2)

Type I B

Group 3:

(4,2)	(2,3)
(3,1)	**(3,4)**

Type II A

Group 4:

(4,3)	(2,2)
(3,1)	**(3,4)**

Type II B 1 a

(4,3)	(2,1)
(3,2)	**(3,4)**

Type II B 1 b

(4,4)	(2,2)
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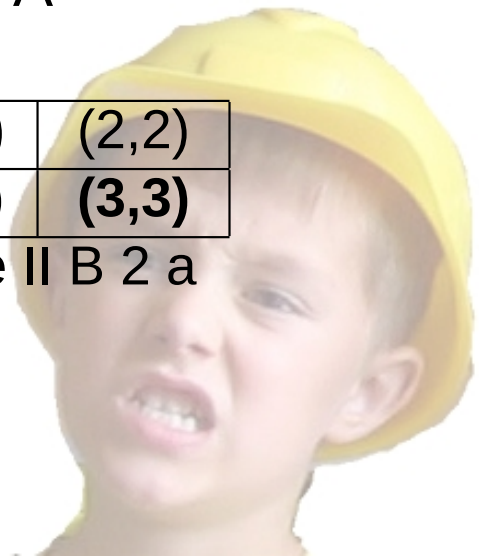
Type II B 2 a

(4,4)	(2,1)
(3,2)	**(3,3)**

Type II B 2 b

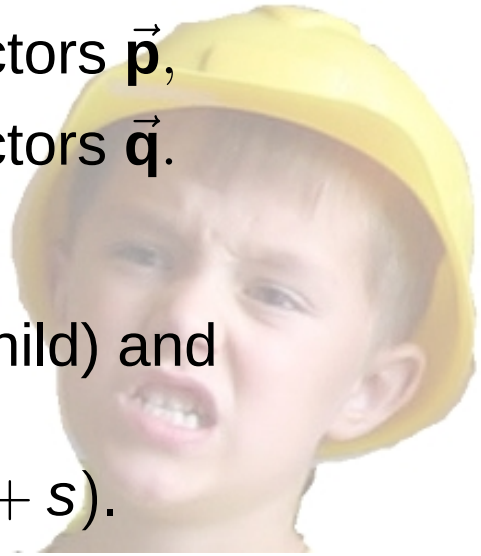


Mixed strategies: Nash equilibria

$$A = \begin{bmatrix} 1 & -s \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1-c & -s \\ -c & 0 \end{bmatrix}.$$

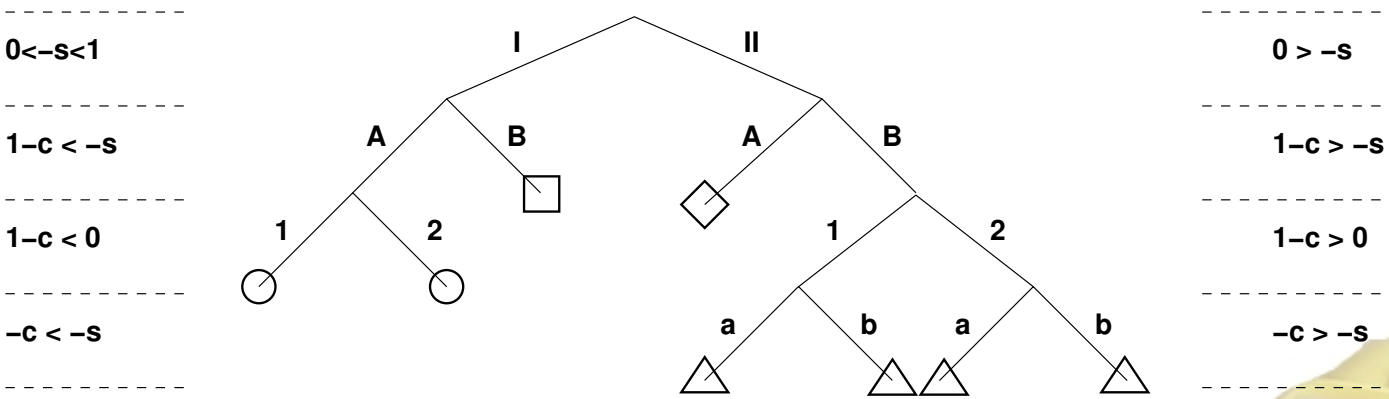
Seek Nash equilibrium $(\vec{\mathbf{p}}^*, \vec{\mathbf{q}}^*)$:

$$\begin{aligned} \vec{\mathbf{p}}^{*\mathsf{T}} A \vec{\mathbf{q}}^* &\geq \vec{\mathbf{p}}^{\mathsf{T}} A \vec{\mathbf{q}}^* && \text{for all probability vectors } \vec{\mathbf{p}}, \\ \vec{\mathbf{p}}^{*\mathsf{T}} B \vec{\mathbf{q}}^* &\geq \vec{\mathbf{p}}^{*\mathsf{T}} B \vec{\mathbf{q}} && \text{for all probability vectors } \vec{\mathbf{q}}. \end{aligned}$$



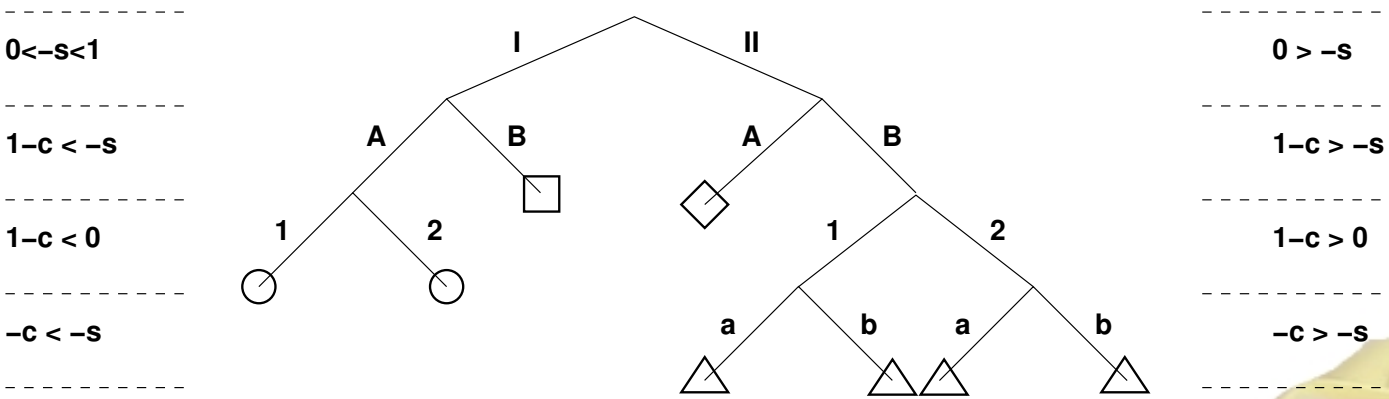
This only applies to Group 4 with $p = c/(1+s)$ (child) and $q = s/(1+s)$ (mother).
The child's payoff is 0 and the mother's is $-cs/(1+s)$.

Grand summary

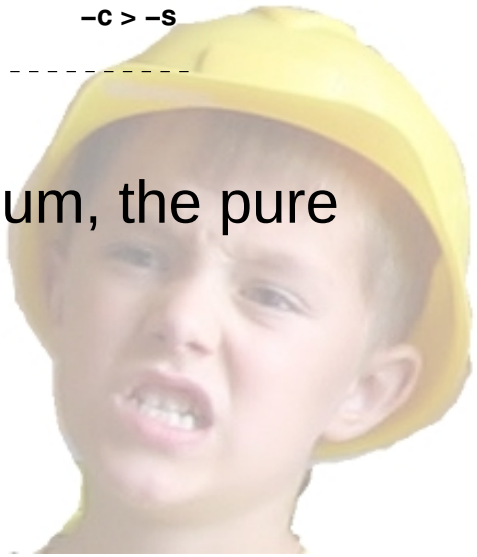


Group 1 (circle): There is only one Nash equilibrium, the pure strategy pair “Go” - “Ignore”. Secure.

Grand summary



Group 2 (square): There is only one Nash equilibrium, the pure strategy pair “Go” - “Attend”. Secure.



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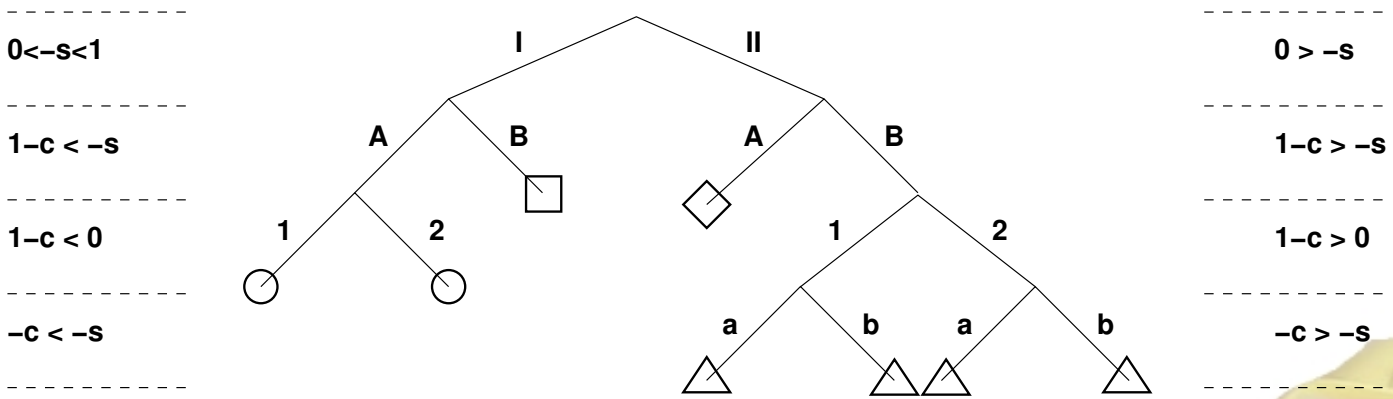
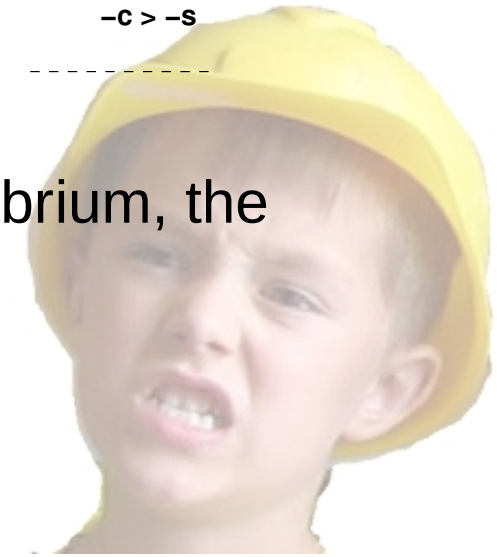


Group 3 (diamond): There is only one Nash equilibrium, the pure strategy pair “Don’t go” - “Ignore”. Avoidant.



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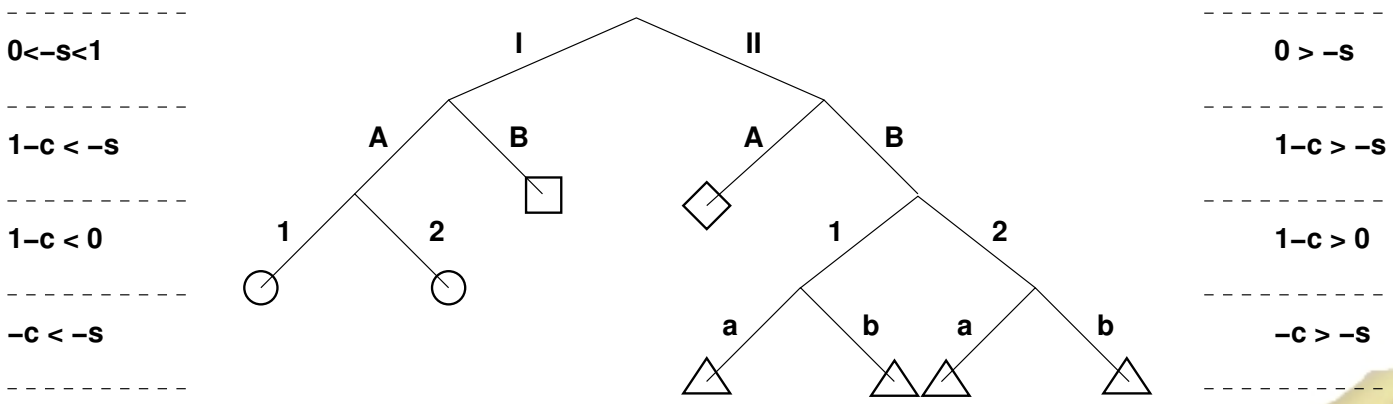
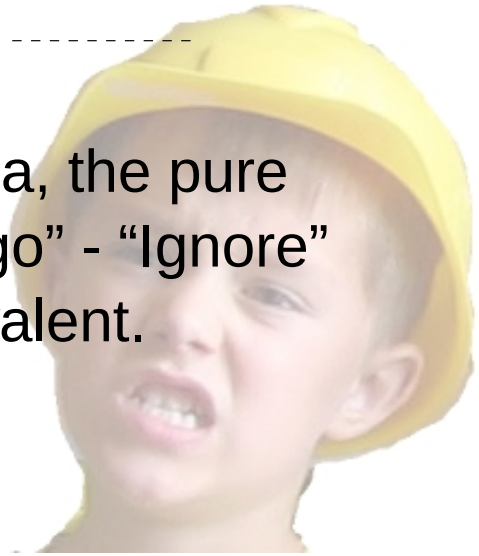
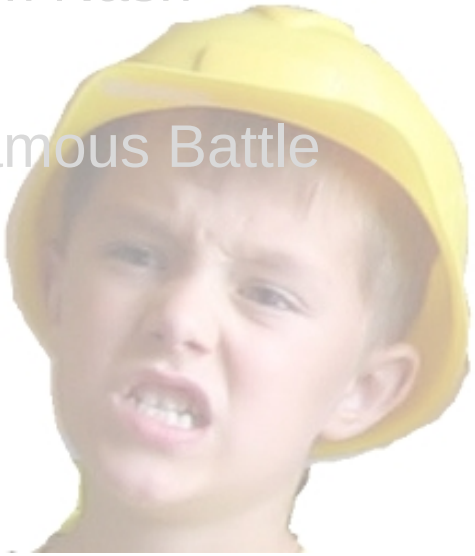


Group 4 (triangles): There are three Nash equilibria, the pure strategy pairs “Go” - “Attend” (secure) and “Don’t go” - “Ignore” (avoidant). The mixed equalizing strategy is ambivalent.



Conclusions

- A small number of parameters characterized a very rich decision structure.
- Seemingly insane behavior in quite rational.
- Counseling could move child-mother pairs from Nash equalibrium to another.
- Group 4 structure can lead to resentment. (Famous Battle of the Sexes.) This has no known resolution.



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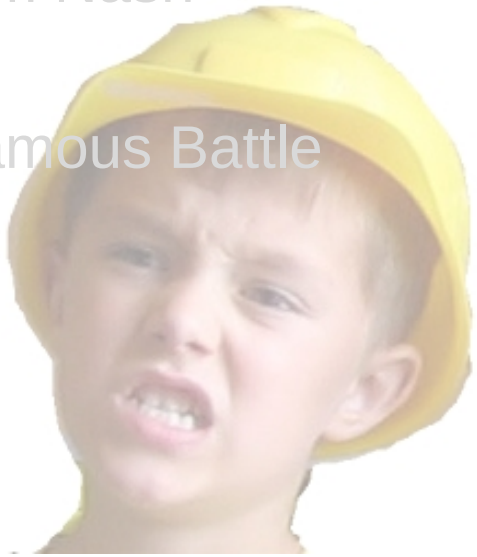

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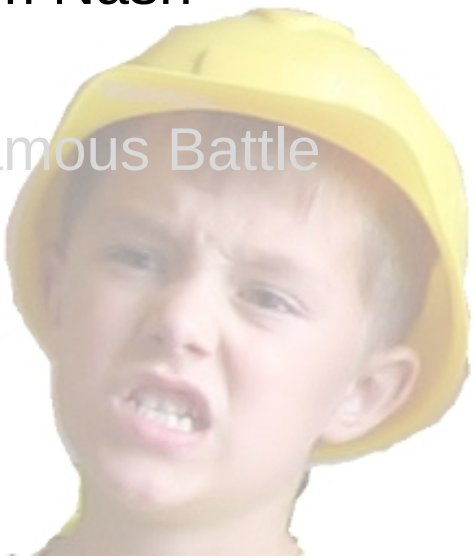

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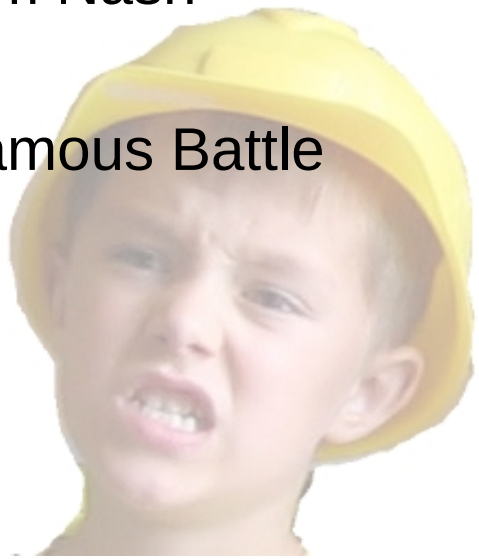
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